Example 2.6

The weights (in kg.) of 8 students of class VII are 38, 42, 43, 50, 48, 45, 52 and 50. Test the hypothesis that the variance of the population is 48 kg, assuming the population is normal and μ is unknown.

Solution:

Step 1 : **Null Hypothesis** H_0: $\sigma^2 = 48$ kg.

i.e. Population variance can be regarded as 48 kg.

Alternative hypothesis H_1: $\sigma^2 \neq 48$ kg.

i.e. Population variance cannot be regarded as 48 kg.

Step 2 : The given sample information is
Sample size $(n) = 8$

Step 3 : **Level of significance**
$\alpha = 5\%$

Step 4 : **Test statistic**

Under null hypothesis H_0

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$$

follows chi-square distribution with $(n-1)$ *d.f.*

Step 5 : Calculation of test statistic

The value of chi-square under H_0 is calculated as under:

To find $\bar{x}$ and sample variance s^2, we form the following table.

x_i	$(x_i - 46)$	$(x_i - 46)^2$
38	-8	64
42	-4	16
43	-3	9
50	4	16
48	2	4
45	-1	1
52	6	36
50	4	16
$\sum_{i=1}^{8} x_i = 368$	0	$\sum_{i=1}^{8} (x_i - \bar{x})^2 = 162$

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